

# ISS Greenhouse Module

Campaign 1 · Case 01 · Low Earth Orbit



SOLAR  
AGRICULTURAL  
AGENCY  
Teacher Guide

## PREPARATION

### Before class

- Confirm the game loads.
- Print the student sheet double-sided or use fillable HTML mode.
- Keep the evidence fallback ready.
- Provide headphones.

## GAME ACCESS

### Launch path

[interstellarharvest.github.io/Space-Sprout-Sleuth/](https://interstellarharvest.github.io/Space-Sprout-Sleuth/)

Select Campaign 1 → Case 1.

## CORRECT DIAGNOSIS

### Central interpretation

Microgravity disrupts gravitropism. Roots need an alternative consistent cue or guidance system.

## ESSENTIAL EVIDENCE

### Evidence students should notice

| Source  | Essential evidence                                                                                              |
|---------|-----------------------------------------------------------------------------------------------------------------|
| Crew    | Roots do not grow down; resource adjustments did not solve the pattern.                                         |
| Sensors | Microgravity is the major unusual condition; nutrients, water, temperature, humidity, and lighting are nominal. |
| Plants  | Dense tangles, poor anchoring, and unusual stem angles.                                                         |
| Logs    | Gravitropism/statolith mechanism and possible replacement cues.                                                 |

## LIKELY STICKING POINT

Microgravity is not the literal absence of gravity. The ISS and everything inside it are continuously falling around Earth, producing apparent weightlessness.

## COLLECTION

### What to collect

3 · Investigate four evidence sources · 6 · Diagnose and reject an alternative · 7 · Claim-Evidence-Reasoning · 8 · Supply a consistent orientation cue · 9 · Exit ticket

## LESSON FLOW

### 60-minute plan

| Time  | Action                                                                |
|-------|-----------------------------------------------------------------------|
| 0–5   | Launch game and materials                                             |
| 5–10  | Initial prediction                                                    |
| 10–30 | 3 · Investigate four evidence sources                                 |
| 30–38 | 4 · Test the competing explanations                                   |
| 38–48 | 6 · Diagnose and reject an alternative + 7 · Claim-Evidence-Reasoning |
| 48–55 | 8 · Supply a consistent orientation cue                               |
| 55–60 | 9 · Exit ticket                                                       |

## FALLBACK

### Technical fallback

Use the evidence digest on the final teacher page. Students complete the same academic task without the game.

## TEACHER LINE

“Your job is not to click everything. Your job is to determine which evidence best explains the root pattern.”

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## OVERVIEW

### Lesson overview

Students investigate a stunted lettuce crop aboard the International Space Station. They compare crew testimony, environmental sensors, direct plant observations, and mission logs to determine why roots form tangled balls despite normal resources. Students explain the gravitropism mechanism and evaluate an engineered alternative cue.

#### GUIDING QUESTION

Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

## ALIGNMENT

### Standards alignment

- **MS-ETS1-1:** Supports defining criteria and constraints for a root-guidance solution.
- **MS-ETS1-2:** Supports evaluating competing orientation solutions.
- **SEP:** Constructing Explanations and Designing Solutions; Analyzing and Interpreting Data.
- **DCI:** ETS1.A and ETS1.B, with plant structure/function as supporting science.
- **CCC:** Cause and Effect; Structure and Function.

This lesson supports components of the standards; it does not by itself fulfill an entire performance expectation.

#### SUCCESS CRITERIA

### What proficient work includes

Relevant evidence from at least three sources · correct mechanism · evidence-based alternative rejection · concise CER · directional engineering response.

## REFERENCE

### Vocabulary

gravitropism · microgravity · statocyte · statolith · phototropism

## OUTCOMES

### Learning objectives

1. Distinguish observations from explanations across four evidence sources.
2. Identify disrupted gravitropism as the best-supported cause.
3. Explain statolith movement as part of a directional gravity-sensing mechanism.
4. Reject an alternative explanation using evidence.
5. Define a criterion and constraint for an orientation system.

## PREPARATION

### Materials

Chromebooks · headphones · game access · double-sided mission sheet or editable HTML · evidence fallback.

#### NOTES

## PREPARATION

### Teacher preparation

1. Open the game and confirm Campaign 1, Case 1 loads.
2. Test audio/headphones.
3. Print or share the student sheet.
4. Keep the evidence fallback ready.

## ☒ PROCEDURE

### 0–5 minutes · Setup

Students launch the game and open the mission sheet. State that game score, rank, speed, and optional dialogue are not academically graded.

## PROCEDURE

### 5–10 minutes · Phenomenon launch

Show the greenhouse image or read the briefing. Ask what evidence would distinguish a resource problem from an orientation problem. Students complete **2 · Initial thinking**; do not confirm the diagnosis.

## PROCEDURE

### 10–30 minutes · Gameplay investigation

Students visit crew, sensors, plants, and logs. They record only evidence that supports or rejects a cause.

## FACILITATION PROMPTS

Which detail is a symptom? Which normal reading rules out an explanation? Have you used more than one source?

## ☒ PROCEDURE

### 30–38 minutes · Evidence checkpoint

Students complete **4 · Test the competing explanations**. Confirm that resource readings are normal and that the root pattern is directional and repeated. Do not require optional dialogue completion.

## PROCEDURE

### 38–48 minutes · Diagnosis and CER

Students complete **6 · Diagnose and reject an alternative** after the game unlocks, then complete **7 · Claim-Evidence-Reasoning** using evidence from several sources and reasoning that links microgravity to a weakened directional cue.

## PROCEDURE

### 48–55 minutes · Engineering application

Students choose an alternative cue and give one criterion and constraint. Strong responses distinguish a direct root cue from lighting that primarily guides shoots.

## ☒ PROCEDURE

### 55–60 minutes · 9 · Exit ticket

Students answer independently. Collect or save the mission file.

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## ✓ FORMATIVE CHECKS

### Checks for understanding

- Separates symptom from cause.
- Uses normal readings to reject alternatives.
- Describes microgravity as apparent weightlessness.
- Links statolith direction to root orientation.
- Proposes a cue that addresses the problem.

## 📖 ASSESSMENT

### Evidence of learning

**Formative:** prediction, evidence matrix, explanation ranking, teacher questions.

**Collected:** diagnosis, rejected alternative, CER, engineering response, exit ticket.

## ♿ ACCESS

### Accessibility supports

- Bullet-point evidence accepted.
- CER frame available but optional.
- Text-to-speech compatible.
- Game evidence may remain open.
- Evidence fallback available.
- No group product required.

## ❓ MISCONCEPTIONS

### Common misconceptions

1. **No gravity:** The ISS remains under gravity while continuously falling around Earth.
2. **Roots use only gravity:** Roots also respond to moisture, touch, chemicals, and other cues.
3. **Brighter light fixes roots:** Light is a stronger shoot cue; roots need a more direct guide.
4. **Plants cannot grow in space:** Growth is possible with adapted environmental systems.
5. **Plant pillows recreate gravity:** They manage media and fluids, not a gravity vector.

## 🧪 SOURCE STATUS

### Science status



#### ESTABLISHED

gravitropism, statolith involvement, phototropism, plant responses to microgravity.



#### PLAUSIBLE ENGINEERING

root channels, moisture cues, rotating systems. Students complete **8 · Supply a consistent orientation cue** on the worksheet; Campaign 1 has no in-game apply-the-fix step, so these solutions appear only as discussion in the mission logs, not as a selectable action.



#### FICTIONAL CONTEXT

the specific emergency, crew, SAA, and exact symptoms.

## 🔑 FALLBACK

### Technical fallback

Distribute the evidence digest. Students complete the same mission sheet; technical failure is not graded.

NOTES

EVIDENCE ARCHITECTURE Evidence map

| Source  | Essential evidence                                              | Role                                              |
|---------|-----------------------------------------------------------------|---------------------------------------------------|
| Crew    | Roots do not grow down; changing resources did not help.        | Symptom and failed interventions                  |
| Sensors | Microgravity; normal nutrients, water, climate, and lighting.   | Strong causal difference and elimination evidence |
| Plants  | Tangled roots, poor anchoring, unusual stems.                   | Direct biological observation                     |
| Logs    | Gravitropism/statolith mechanism and possible replacement cues. | Mechanism and design options                      |

REASONING PATH How the case resolves

1. Root tangling is the symptom.
2. Normal resources weaken simple resource explanations.
3. Microgravity is the major unusual environmental condition.
4. Logs connect gravity sensing to directional growth.
5. Disrupted gravitropism best explains the pattern.
6. A useful solution supplies another consistent cue.

DISTRACTORS Competing explanations

| Distractor        | Why tempting              | Why rejected                                                                        |
|-------------------|---------------------------|-------------------------------------------------------------------------------------|
| Nutrient burn     | Pale, stunted plants      | Nominal solution; directional tangling is a poor match                              |
| Wrong light cycle | Odd stems and pale leaves | 16/8 and PAR are nominal; roots are the strongest abnormality                       |
| Defective seeds   | All plants affected       | Pattern repeats with environment and prior plantings                                |
| Water problem     | Roots seek water          | Delivery works; uniform wicking may reduce a cue but does not best explain the case |

The nutrient, light, and seed distractors are the three incorrect options students can select on the game's Diagnose screen. "Water problem" is an additional teacher discussion distractor for class conversation — it is not one of the selectable in-game diagnoses.

BOUNDARY NOTE

The game simplifies a complex plant response. Do not teach that roots are completely random, that statoliths are the only gravity-sensing mechanism, or that plant pillows fully solve orientation.

INSTRUCTIONAL IDENTITY Student role and primary practice

**Student role:** Pattern investigator. **Primary practice:** compare multiple evidence types, eliminate alternatives, explain a mechanism, and define an engineering problem.

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## ☒ CHECKLIST RUBRIC

### Quick classroom rubric

| Dimension     | Secure                                                           | Developing                                              | Beginning                             |
|---------------|------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------|
| Evidence      | Specific, relevant evidence from several sources                 | Some relevant evidence; important detail/source missing | Vague, copied, or irrelevant evidence |
| Mechanism     | Connects microgravity, statolith direction, and root orientation | Names gravitropism but mechanism is incomplete          | Cause stated without mechanism        |
| Reasoning     | Connects evidence to claim and rejects an alternative            | Partial connection or weak rejection                    | Evidence listed without explanation   |
| Communication | Accurate vocabulary and complete product                         | Minor vocabulary/completion gaps                        | Major gaps impede meaning             |

## ☐ ANALYTIC GRID

### Formal analytic rubric

| Criterion              | 4 · Accomplished                           | 3 · Proficient                           | 2 · Developing                  | 1 · Beginning                |
|------------------------|--------------------------------------------|------------------------------------------|---------------------------------|------------------------------|
| Evidence selection     | Most relevant; symptom/cause distinguished | Relevant with minor omissions            | Mixed relevant/irrelevant       | Little usable evidence       |
| Source diversity       | Integrates 3–4 source types                | Uses at least 2 source types             | Mostly one source               | Sources unclear              |
| Mechanism              | Accurate and appropriately qualified       | Central mechanism accurate               | Important gaps                  | Unsupported/inaccurate       |
| Alternative evaluation | Specific contradictory evidence            | Generally relevant evidence              | Weak support                    | No evaluation                |
| Engineering            | Feasible cue, criterion, constraint        | Relevant solution + criterion/constraint | Loosely related solution        | Does not address orientation |
| Communication          | Clear, concise, complete                   | Mostly clear/complete                    | Inconsistent but understandable | Incomplete/difficult         |

**SCORING BOUNDARY**  
No numeric points appear on the student page by default. Game score, rank, speed, and optional exploration are never academic criteria.

#### NOTES

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## AUTHORITATIVE SOURCES

### Source list

1. NASA, *Growing Plants in Space*  
[nasa.gov/exploration-research-and-technology/growing-plants-in-space/](https://nasa.gov/exploration-research-and-technology/growing-plants-in-space/)
2. NASA Ames, *Plant Gravity Perception (Space-13)*  
[nasa.gov/ames/space-biosciences/plant-gravity-perception-spacex-13/](https://nasa.gov/ames/space-biosciences/plant-gravity-perception-spacex-13/)
3. NASA, *Veggie Plant Growth System Activated on ISS*  
[nasa.gov/missions/station/veggie-plant-growth-system-activated-on-international-space-station/](https://nasa.gov/missions/station/veggie-plant-growth-system-activated-on-international-space-station/)
4. NASA Science, *Plant Biology Program · Hardware*  
[science.nasa.gov/biological-physical/focus-areas/plant-biology/hardware/](https://science.nasa.gov/biological-physical/focus-areas/plant-biology/hardware/)
5. Ditengou, Teale & Palme (2020), *Settling for Less: Do Statoliths Modulate Gravity Perception?*  
[pubmed.ncbi.nlm.nih.gov/31963631/](https://pubmed.ncbi.nlm.nih.gov/31963631/)
6. NASA OSDR, *Arabidopsis Seedlings in Microgravity (OSD-38)*  
[osdr.nasa.gov/bio/repo/data/studies/OSD-38](https://osdr.nasa.gov/bio/repo/data/studies/OSD-38)
7. NGSS MS-ETS1-1 and MS-ETS1-2  
[nextgenscience.org](https://nextgenscience.org)

## EVIDENCE FALLBACK

### No-game evidence digest

**Briefing:** ISS lettuce is stunted; roots curl around plant pillows instead of anchoring.

**Crew:** roots do not grow downward; changing resources did not help.

**Sensors:** microgravity; nutrients, water, temperature, humidity, and lighting are nominal.

**Plants:** dense tangles, poor anchoring, and angled stems.

**Logs:** statolith-based gravity sensing normally provides direction; possible compensations include root channels, moisture gradients, shoot lighting, or rotating systems.

## TECHNICAL NOTES

### Classroom fallback

- Refresh once if the game does not load.
- Use a current Chromebook Chrome tab.
- The Diagnose screen unlocks after formal clues are collected.
- Do not require every optional conversation branch.
- Move to the fallback within two minutes rather than losing lesson time.

## NOTES